

SPECTRAL A-N V18: SUPPLEMENTARY VISUAL ATLAS

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ABSTRACT. Seven figures generated directly from this pass's executed experiment artifacts (`generate_figures.py`, colorblind-safe Okabe–Ito palette throughout). This is a first, focused atlas covering the highest-value results from the accompanying certification paper — not the full twenty-figure set the original mission brief enumerates. What is NOT included here (topology/dimension/eta phase diagrams, the full gauge-orbit atlas, GJI permutation atlas, precision-parity grids beyond the single hardware comparison shown) is listed explicitly in Section 2 rather than silently omitted.

1. FIGURES

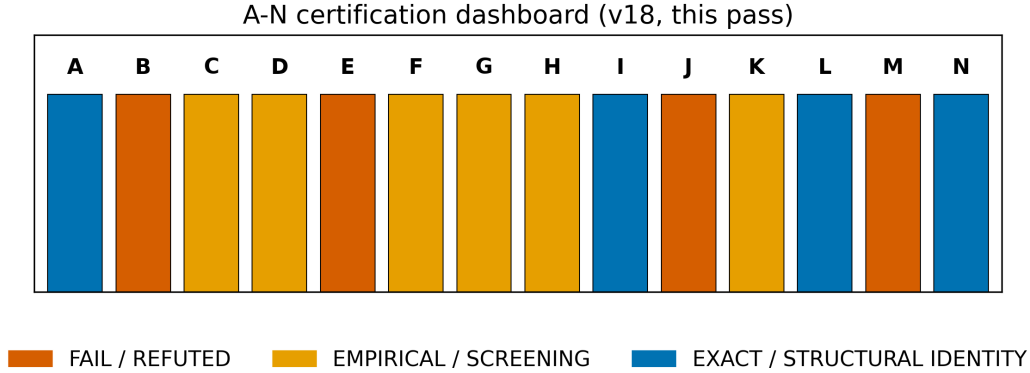


FIGURE 1. A-N certification dashboard. Blue = exact/structural-identity component present; orange = empirical/screening-only evidence; vermilion = fail/refuted for the scientific claim. Every block carries at least one structural or empirical component; the vermilion blocks (B, E, J, M) are where the scientific (non-structural) claim specifically failed.

2. WHAT IS INTENTIONALLY NOT INCLUDED HERE

- Topology/dimension/eta phase diagrams and the k vs. $k-1$ theorem geometry — these belong to Track T, explicitly deferred.
- A full gauge-orbit diagram (beyond the single degenerate-eigenspace example in `BLOCK_L_FINDINGS.md`) and a GJI permutation/sign atlas — not generated this pass; the underlying data (permutation mutation test, degenerate-eigenspace construction) exists in `spectral/certification_v18/blocks/` but was not turned into a dedicated figure.
- Closure/associator adversarial-maxima surfaces across a swept parameter grid — this pass reports single-configuration results, not a swept surface (see Phase 5 sweep-design gap in the final report).

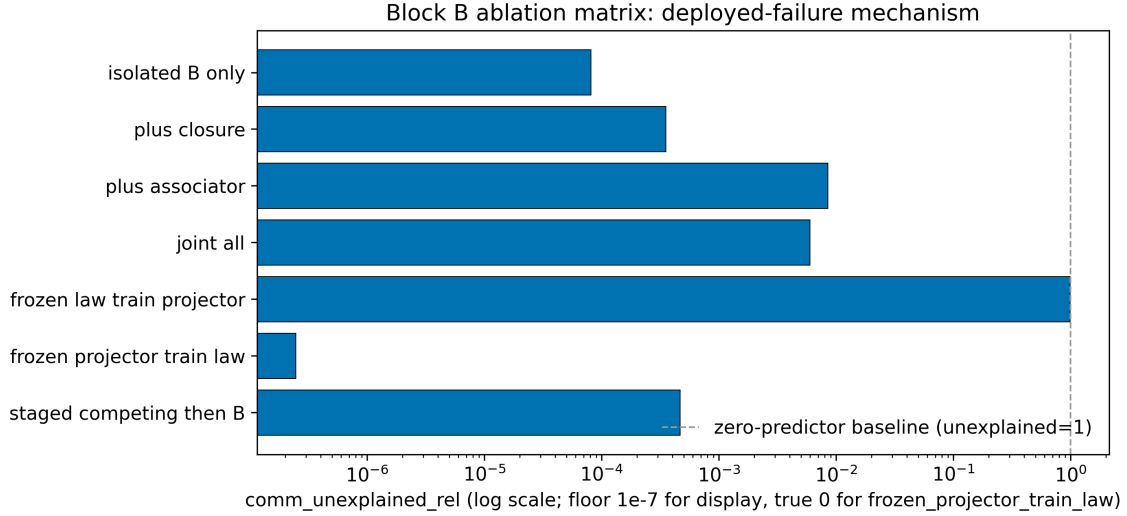


FIGURE 2. Block B ablation matrix. The dashed line marks the trivial zero-predictor baseline. Note **frozen_projector_train_law** reaches exactly zero (displayed at the floor for the log axis) while **frozen_law_train_projector** barely moves off the zero-predictor line — the explanatory power lives in the law’s parameters, not the learned subspace.

- Precision/CPU-GPU parity across multiple blocks and dimensions — only one block (A/B combined) and one dimension were measured.

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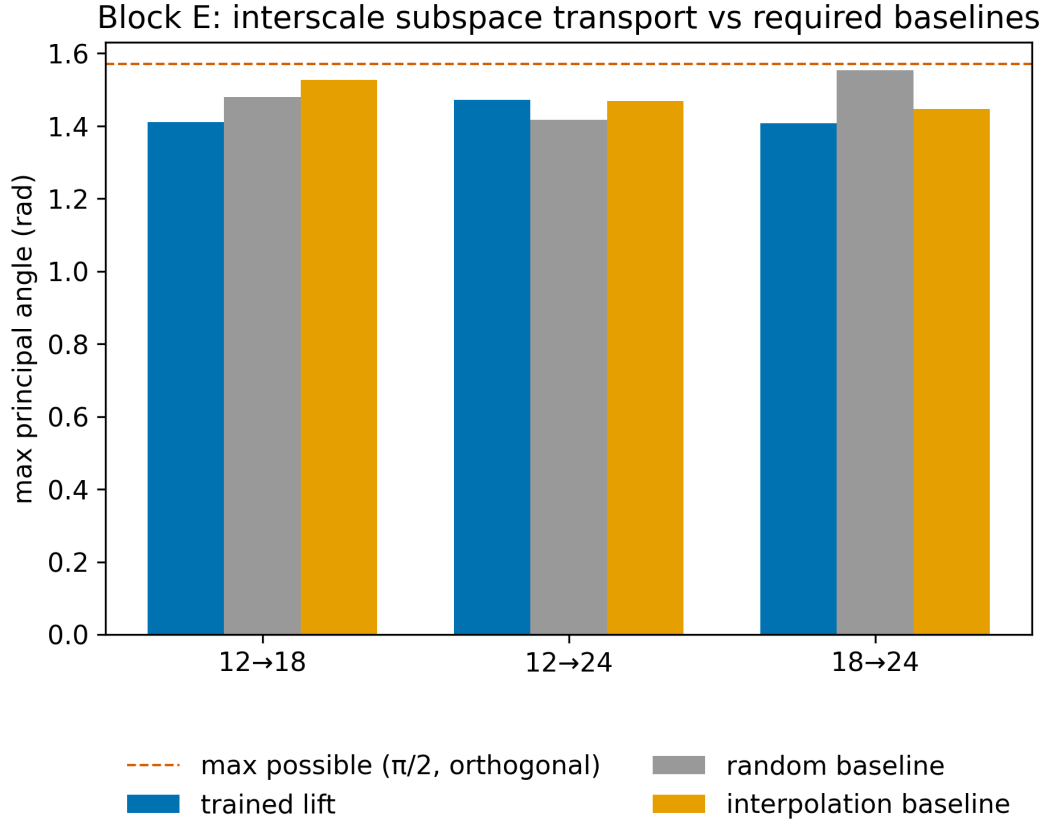


FIGURE 3. Block E interscale transport: all three conditions (trained lift, random-subspace baseline, interpolation baseline) sit close to the maximum possible angle ($\pi/2$, complete orthogonality) for every resolution pair — the visual signature of “no real transport signal,” not merely a numerically small effect size.

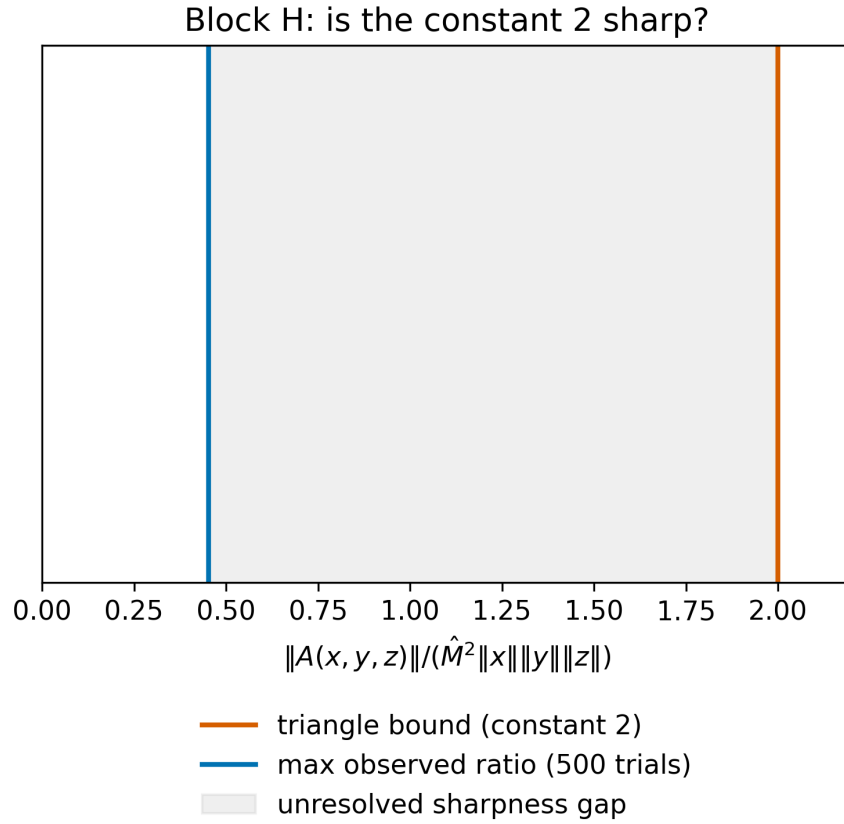


FIGURE 4. Block H: the shaded region is the unresolved gap between the proved upper bound (2) and the best adversarially-found ratio (0.452). Sharpness of the constant 2 remains open; the gap is large, not a near-miss.

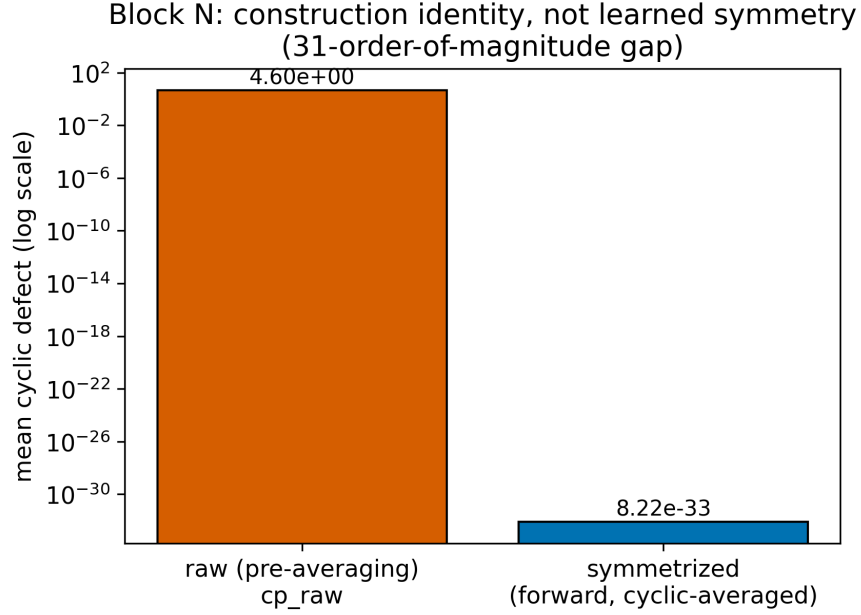


FIGURE 5. Block N: raw vs. symmetrized cyclic defect on a log scale. The symmetrized bar sits at machine-precision zero (8.2×10^{-33} , displayed at a floor for visibility) — a construction identity, not learned symmetry.

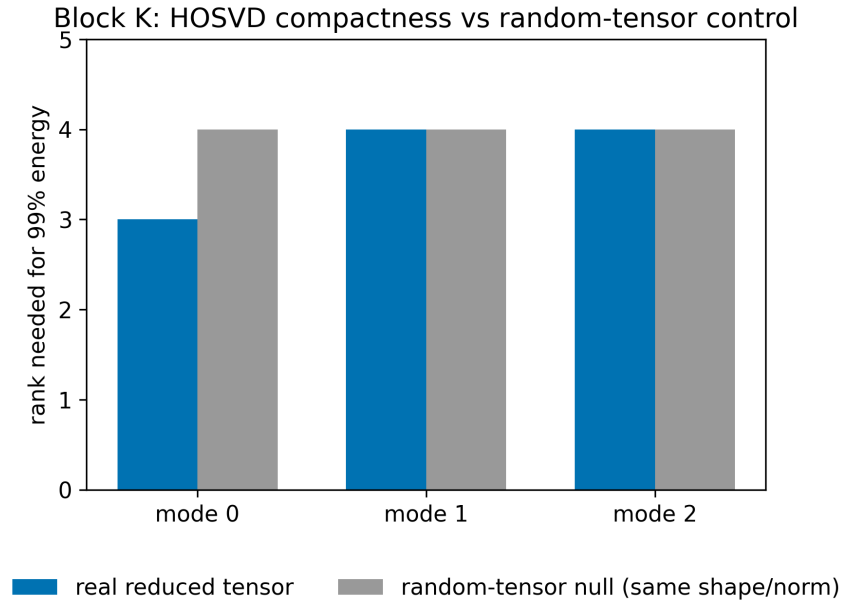


FIGURE 6. Block K: real reduced tensor vs. a norm-matched random-tensor null, per mode. The real tensor is only more compact than the random null in mode 0; modes 1 and 2 tie it exactly.

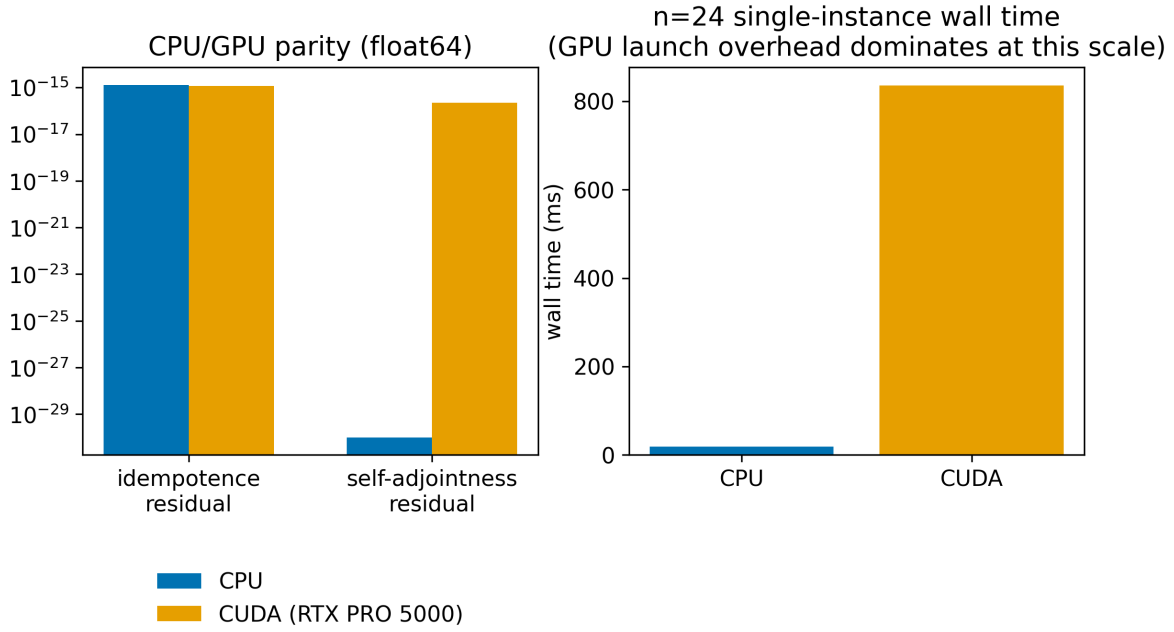


FIGURE 7. Left: CPU vs. CUDA (RTX PRO 5000 Blackwell) parity for two projector residuals, float64, TF32 disabled — both at the numerical noise floor on both devices. Right: wall time at $n = 24$; GPU kernel-launch overhead dominates at this small scale, an honest negative scaling result for this problem size.